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-- Data Quality Audit
SELECT
    COUNT(*) AS total_visits,
    SUM(CASE WHEN checkout_time IS NULL THEN 1 ELSE 0 END) AS ghost_sessions,
    SUM(CASE WHEN consult_start < triage_start THEN 1 ELSE 0 END) AS clock_sync_errors,
    SUM(CASE WHEN lab_start IS NULL THEN 1 ELSE 0 END) AS no_lab_required,
    MIN(checkin_time) AS earliest_visit,
    MAX(checkin_time) AS latest_visit
FROM emr_clinic_visits;
```

Analysis

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-- 1. Branch x Month Trend
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```
SELECT
    o.branch,
    o.visit_month,
    COUNT(DISTINCT o.visit_id) AS total_visits,
    ROUND(AVG(o.total_turnaround_mins), 1) AS avg_total_tat_mins,
    COUNT(n.nps_score) AS nps_responses,
    ROUND(AVG(n.nps_score), 1) AS avg_nps_score,
    ROUND(
        (SUM(CASE WHEN n.nps_score >= 9 THEN 1 ELSE 0 END) -
         SUM(CASE WHEN n.nps_score <= 6 THEN 1 ELSE 0 END))
        * 100.0 / NULLIF(COUNT(n.nps_score), 0), 1
    ) AS calculated_nps
FROM v_operational_metrics o
LEFT JOIN sms_nps_feedback n ON o.visit_id = n.visit_id
GROUP BY o.branch, o.visit_month
ORDER BY o.branch, o.visit_month;
```

branch	visit_month	total_visits	avg_total_tat_mins	nps_responses	avg_nps_score	calculated_nps
Kawangware	2026-03	985	45.4	372	7.4	6.7
Kawangware	2026-04	905	57.1	352	5.7	-39.8
Kawangware	2026-05	926	72.1	352	3.9	-83.8
Sunton	2026-03	497	45.3	213	7.3	6.1
Sunton	2026-04	504	45.0	206	7.4	11.7
Sunton	2026-05	477	45.1	221	7.4	12.2
Umoja	2026-03	965	46.0	385	7.1	1.3
Umoja	2026-04	982	58.4	403	5.9	-39.0
Umoja	2026-05	891	72.5	386	4.0	-81.9

-- 2. Stage-by-Stage Bottleneck Breakdown

```
SELECT
    branch,
    visit_month,
    ROUND(AVG(triage_wait_mins), 1) AS avg_triage_wait,
    ROUND(AVG(consult_wait_mins), 1) AS avg_consult_wait,
    ROUND(AVG(lab_wait_mins), 1) AS avg_lab_wait,
    ROUND(AVG(lab_and_pharm_wait_mins), 1) AS avg_lab_pharm_wait,
    ROUND(AVG(pharmacy_to_exit_mins), 1) AS avg_pharmacy_exit_wait,
    ROUND(AVG(total_turnaround_mins), 1) AS avg_total_tat
FROM v_penda_operational_metrics
GROUP BY branch, visit_month
ORDER BY branch, visit_month;
```

branch	visit_month	avg_triage_wait	avg_consult_wait	avg_lab_wait	avg_lab_pharm_wait	avg_pharmacy_exit_wait	avg_total_tat
Kawangware	2026-03	11.1	7.5	14.6	22.2	4.5	45.4
Kawangware	2026-04	11.3	7.4	14.6	34.0	4.5	57.1
Kawangware	2026-05	11.2	7.6	14.5	48.9	4.5	72.1
Sunton	2026-03	12.0	7.5	14.9	21.3	4.5	45.3
Sunton	2026-04	11.4	7.7	14.5	21.4	4.5	45.0
Sunton	2026-05	11.9	7.6	14.7	21.3	4.4	45.1
Umoja	2026-03	11.0	7.6	14.4	22.9	4.5	46.0
Umoja	2026-04	11.8	7.5	14.5	34.5	4.5	58.4
Umoja	2026-05	11.0	7.5	14.5	49.4	4.5	72.5

-- 3. Patient Abandonment Rate by Branch x Month

```
SELECT
    branch,
    DATE_FORMAT(checkin_time, '%Y-%m') AS visit_month,
    COUNT(*) AS total_visits,
    SUM(CASE WHEN checkout_time IS NULL THEN 1 ELSE 0 END) AS abandoned_visits,
    ROUND(SUM(CASE WHEN checkout_time IS NULL THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS abandonment_rate_pct
FROM emr_clinic_visits
GROUP BY branch, visit_month
ORDER BY branch, visit_month;
```

branch	visit_month	total_visits	abandoned_visits	abandonment_rate_pct
Kawangware	2026-03	1030	19	1.84
Kawangware	2026-04	947	23	2.43
Kawangware	2026-05	1004	57	5.68
Sunton	2026-03	527	14	2.66
Sunton	2026-04	528	8	1.52
Sunton	2026-05	491	5	1.02
Umoja	2026-03	1001	20	2.00
Umoja	2026-04	1027	30	2.92
Umoja	2026-05	945	43	4.55

-- 4. NPS Tier Breakdown by Branch (full population)

SELECT

e.branch,
COUNT(CASE WHEN n.nps_score >= 9 THEN 1 END) AS promoters,
COUNT(CASE WHEN n.nps_score BETWEEN 7 AND 8 THEN 1 END) AS passives,
COUNT(CASE WHEN n.nps_score <= 6 THEN 1 END) AS detractors,
COUNT(n.nps_score) AS total_responses,
ROUND(
 (COUNT(CASE WHEN n.nps_score >= 9 THEN 1 END) -
 COUNT(CASE WHEN n.nps_score <= 6 THEN 1 END))
 * 100.0 / COUNT(n.nps_score), 1
) AS calculated_nps

FROM sms_nps_feedback n

JOIN emr_clinic_visits e ON n.visit_id = e.visit_id

GROUP BY e.branch

ORDER BY calculated_nps;

branch	promoters	passives	detractors	total_responses	calculated_nps
Umoja	228	254	729	1211	-41.4
Kawangware	209	265	663	1137	-39.9
Sunton	252	219	198	669	8.1